

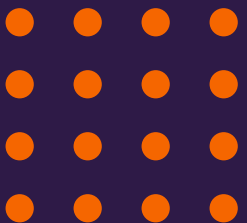


Unsupervised Learning – Clustering

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Unsupervised Learning Basics

- We want to discover subgroups among variables or observations, but not in how they related to a response variable
- There's usually no precise goal of unsupervised clustering; it's often used as a descriptive tool
- It's easier to obtain data without a response variable (i.e., label)
- Example Applications
 - Find subgroups of breast cancer patients with similar gene expressions
 - Determine subgroups of customers based on browsing and purchasing patterns
 - Group movies according to ratings assigned by viewers

Clustering

- Clustering analysis consists of techniques for finding *subgroups* or *clusters* in a multivariate data set.
- We want to partition the data into distinct groups so that observations within each group are similar to each other
- Need to define concretely what it means for observations to be *similar* or different, which can be domain specific
- Notation
 - Assume n observations are measured on p features
 - Let X be an $n \times p$ matrix of data and X_{ij} for $i = 1, 2, \dots, n$ and $j = 1, \dots, p$
 - We are interested in finding subsets of rows in X that are “similar” to each other, which will form a “cluster”



Simple Example: Eruptions of Old Faithful

Data were collected on time between eruptions and the duration of eruptions for the Old Faithful geyser at Yellowstone National Park

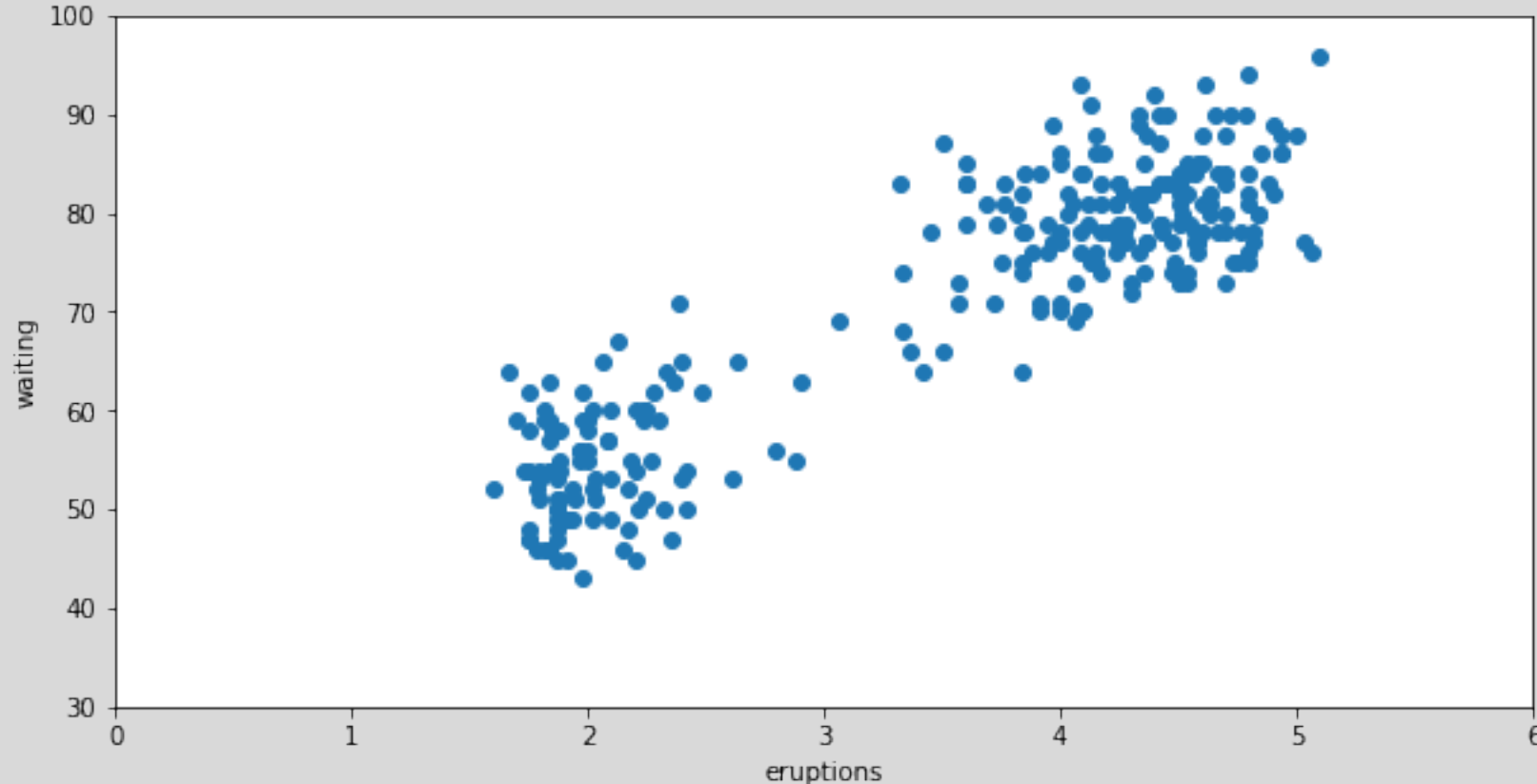
	eruptions	waiting
0	3.600	79
1	1.800	54
2	3.333	74
3	2.283	62
4	4.533	85

	eruptions	waiting
count	272.000000	272.000000
mean	3.487783	70.897059
std	1.141371	13.594974
min	1.600000	43.000000
25%	2.162750	58.000000
50%	4.000000	76.000000
75%	4.454250	82.000000
max	5.100000	96.000000



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